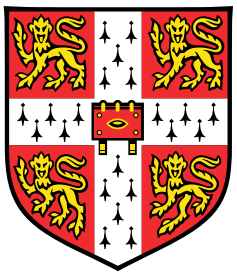
Measure of MightyPix Time Resolution

from Test Beam Data



8275R

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Except where specific reference is made to the work of others, this work is original and has not been already submitted either wholly or in part to satisfy any degree requirement at this or any other university

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Abstract

To meet the demands of increased luminosity at the High-Luminosity Large Hadron Collider (HL-LHC) in 2036, new generations of silicon detectors must achieve higher spatial and timing resolutions and improved radiation tolerance. As part of the LHCb Upgrade II, the Mighty Tracker system will implement High-Voltage Monolithic Active Pixel Sensors (HV-MAPS) based on Complementary Metal-Oxide-Semiconductor (CMOS) technology. MightyPix, a candidate HV-MAPS sensor for this upgrade, is currently in limited supply. Therefore, this project investigates the performance of TelePix2 - a sister sensor with similar architecture and broader availability.

The experiment was performed at the DESY II facility in March, 2024. Test beam data were collected with a setup that included three telescope planes, a scintillator trigger, and a 4 GeV electron beam. Event reconstruction was performed using the Corryvreckan framework.

Time-over-threshold (ToT) and cluster size distributions were analysed across runs with threshold from 140-180 DAC and bias voltages 40-85 V on pixel, 120-130 V on chip. The ToT peak increased with bias voltage and decreased with threshold, consistent with expectations. The mean cluster size exhibits a minimum that shifts to higher threshold values for increasing bias voltage. This behaviour can be explained by the interplay between sensor characteristics and the readout system.

Based on efficiency analysis, the optimal configuration was identified, and under these conditions (130 V chip bias, 140 DAC threshold), the time resolution of TelePix2 was measured to be 2.656 ± 0.003 ns.

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Chapter 1

Introduction

Experimental particle physics today focuses on observing interactions between elementary particles, most of which are well-described by the Standard Model. However, certain observations remain unexplained and point towards new physics beyond the Standard Model - one of the primary motivations driving current research in the field.

To explore these phenomena, experiments require high-energy particles and large statistics, which in turn demand large particle colliders with high luminosity and highly sensitive detectors. The Large Hadron Collider (LHC) is one such facility[1]. Large-scale projects like the LHC often takes years or even decades to plan and build. During this time, new technologies may be developed with improved performance. The silicon sensors for particle tracking is an example of such technological evolution[2].

The current implementation of silicon sensors at the LHC uses hybrid sensors, where the sensing element and the readout electronics are fabricated on separate chips. One major drawback of this approach is the need for fine-pitch bump bonding in pixelated sensors, which significantly adds to both the material and cost budgets. A new sensor concept called high-voltage monolithic active-pixel sensor (HV-MAPS) in complementary metal-oxide-semiconductor (CMOS) technology, was proposed by Ivan Perić in 2007[3]. Since then, several families of sensors have been developed based on this concept[4][5][6].

This project aims to contribute to one of the target applications of HV-MAPS: the pixelated area on the Mighty Tracker of the LHCb detector, at the LHC[7]. Specifically, the time resolution of the target sensor, MightyPix[8], was studied through the test beam performance of a sister sensor, TelePix2[9].

Chapter 2 introduces the Mighty Tracker in the context of the LHC and the LHCb experiment, providing the background and motivation for the development of MightyPix. The theoretical foundation of CMOS HV-MAPS is presented in Chapter 3.

Chapter 4 outlines the test beam setup and describes the methods used for event reconstruction using the Corryvreckan framework[10]. The results of the analysis are summarised and discussed in Chapter 5, followed by the overall conclusions in Chapter 6.

Chapter 2

The LHC

This chapter provides the background of the project, with a focus on the MightyPix sensor, which is designed for use in the Mighty Tracker (MT) sub-detector of the LHCb experiment at the LHC. It introduces the LHC, the LHCb experiment, and the MT system to establish the motivation and application of MightyPix.

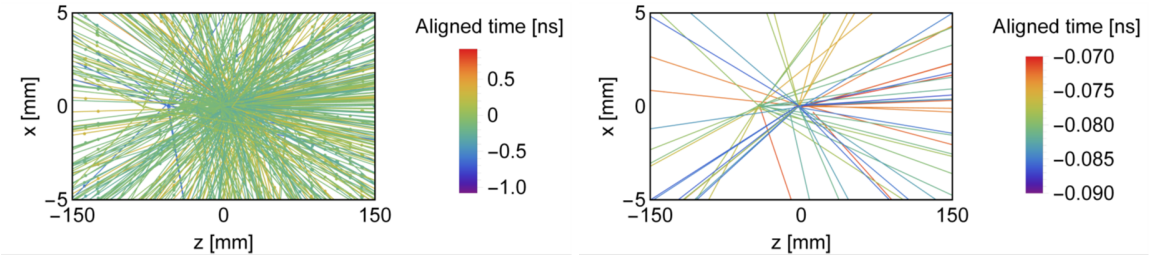
2.1 The Large Hadron Collider

The Large Hadron Collider (LHC) is by far the most powerful particle accelerator in the world[11]. Constructed between 1998 and 2008, it is located 100 metres underground beneath the France–Switzerland border near Geneva. The LHC collides protons at a maximum collision energy of 13TeV, enabling exploration of a broad range of particle physics phenomena, including the study of supersymmetry, the matter-antimatter imbalance, dark matter and dark energy. It also facilitates heavy-ion collisions, providing a platform for investigating the quark-gluon plasma.

At the moment, the precision of the measurements at the LHC is limited by statistics. In other words, the accuracy is limited by finite data size, rather than instrumental limitations or systematic uncertainties. A proportional increase in data size can be achieved either by extending correspondingly the runtime of the experiment, or by increasing the luminosity of the beam. The luminosity measures the amount of particles passing through a given area per unit time in the beam - higher luminosity, more potential particle collisions.

The High-Luminosity Large Hadron Collider project (HL-LHC) plans to increase the working luminosity to $5 \times 10^{34} cm^{-2} s^{-1}$ (about 5 times the current value), with annual data output of $250 fb^{-1}$ and $3000 fb^{-1}$ in 12 years after the upgrade[12]. This would improve the performance of the current collider, exploiting its physics potential until the end of its life span[11].

High luminosity beam puts forward challenges for the detectors at the LHC. Larger number of proton-proton collision per bunch crossing[1] leads to a corresponding increase in the number of primary vertices[2], causing "pile-up" - the overlaying of signals from different primary vertices at a single bunch crossing. This significantly impacts the performance of the detectors. For example, for the MT in LHCb, the efficiency will degrade from 86% to 60%, the ghost rate will increase from 20% to 100%, and the reconstruction time will be doubled[7], if the current sensors are not upgraded. Thus, maintaining the current physical performance will require improvements to detectors' spatial and timing resolution.



(a) time window = 1ns.

(b) time window = 20ps.

Fig. 2.1 Simulation of track density at VELO[7], generated by 42 collisions over one bunch crossing. Applying (b) a 20ps time cut to (a) the bunch crossing time period leaves only a few collisions and tracks. Among all LHCb subdetectors, VELO is the closest to the interaction point, and thus requires highest time resolution - see Fig. 2.2.

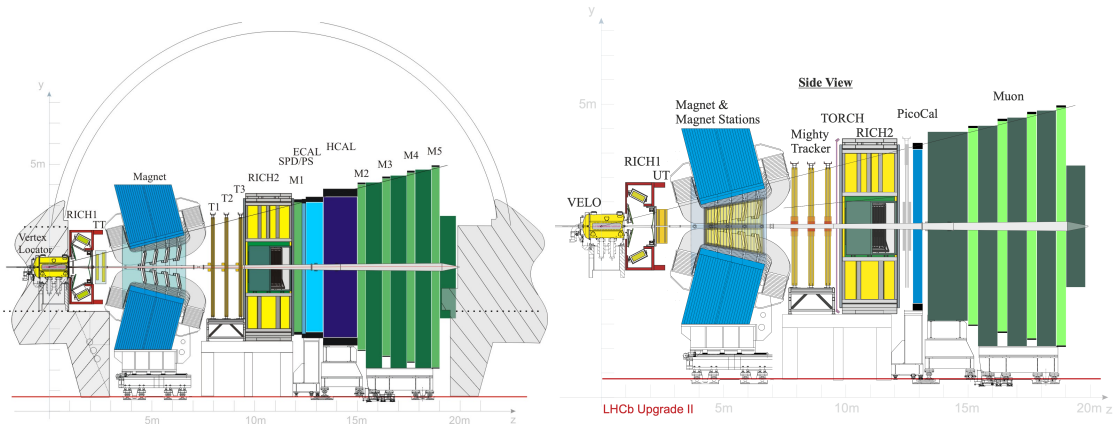
2.2 The LHCb Experiment

Up to the present time, there are nine experiments running at the LHC, each characterised by their detectors. The Large Hadron Collider beauty (LHCb) is the one that focuses on the study of the beauty quark (also known as the bottom quark) and aims to investigate the difference between matter and antimatter.

In LHCb, data is collected from a series of subdetectors, each specialised to a slightly different function. These subdetectors stack behind each other layer-by-layer, and stretches over 20 metres from the collision point along the beam line. They are made to detect mainly the forward particles, as the beauty and charm hadrons and their decay products stay close to the beam line. This design makes LHCb a single forward-arm detector.

[1] At the LHC, particles are accelerated in groups known as "bunches", with a time spacing of 25ns between successive bunches[12]. A bunch crossing occurs when two bunches (one from each beam) pass through each other at an interaction point, and particle collisions can happen.

[2] In one bunch crossing, multiple collisions can occur - between the protons, the products of pp collisions, etc. The protons are called the primary particles, as they start the collision event. A primary vertex is where the primary particles collide - it marks the highest-energy and most significant collisions.



(a) before Upgrade II.

(b) after Upgrade II.

Fig. 2.2 The structure of the LHCb spectrometer before and after Upgrade II[7]. The subdetectors are labelled. During Upgrade II, the sensors will be upgraded, and magnet stations will be added to allow sensing in the magnet section.

As a response to the higher-luminosity challenges from HL-LHC, the LHCb Upgrade II aims to improve the performance of the LHCb detector, allowing experiments to be performed under 7.5× higher pile-up value[7]. This will be achieved by several methods, including the extensive use of MAPS technologies. The existing arrangement of the spectrometer and the majority of subdetectors will be kept, while certain components will be replaced.

2.3 The Mighty Tracker

The MT is the fourth subdetector of the LHCb detector, as shown in Fig.2.2. It consists of three tracking stations, each is divided into an outer and an inner tracker region. The outer region consists of four scintillating fibres (SciFi) detection planes, and each plane contains up to six modules of densely packed fibres. The inner region is equipped with two silicon panels, instrumented with high granularity silicon pixel sensors. A silicon panel is divided into 28 modules, each covers $20\,cm \times 54\,cm$. The beam passes through the centre of the tracking stations.

To meet the requirement of recovering pre-upgrade performance, a mixed technology solution is adopted. For the central silicon pixel region, HV-MAPS produced by a commercial 180nm CMOS technology, are chosen for their radiation tolerance, capability of high-occupancy conditions, and lower costs compared to hybrid sensors. It has been widely used in all range of imaging-device applications, from low-end web-cams to high-end digital cameras[13]. For it to be used in scientific applications such as particle sensing, it must be

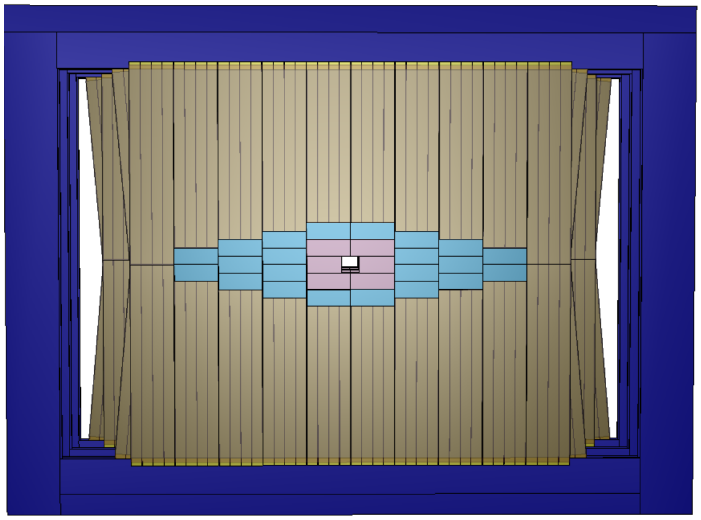


Fig. 2.3 The front view of the MT stations[7]. The pink and light blue regions are covered with silicon sensors. The grey outer layer consists of 4 SciFi planes, two oriented vertically and two tilted by $\pm 5^{\circ}$ with respect to the vertical direction.

tailored to meet the requirement of low noise, high speed, and radiation hardness. This will be discussed in details in Sec. 3.2.

With a bunch spacing of 25 ns[12], one bunch crossing is expected every 25 ns. Assuming a normal distribution of the hits, they need to be assigned correctly to the bunch crossing within 4σ, which means that 99.99% of hits lies within 25 ns window. This requires a σ <3 ns, setting an requirement on sensor time resolution.

Chapter 3

Silicon-based Sensor Technology

In this chapter, the High-Voltage Monolithic Active Pixel Sensors (HV-MAPS) are introduced as a particle sensing technology. The discussion begins with the fundamentals of semiconductors, followed by the principles and the electronics of HV-MAPS. The chapter then introduces the MightyPix and TelePix2 sensors - the latter being studied in details in this project.

3.1 Silicon Semiconductor in Particle Sensing

3.1.1 Silicon as a semiconductor

A semiconductor is a material with electrical conductivity between that of a conductor and an insulator. In conductors, the Fermi level lies within a partially-filled band. In contrast, in semiconductors or insulators, it lies within the band gap between the filled valence band and the vacant conduction band. The band gap E_G is given by

$$E_G = E_C - E_V, \tag{3.1}$$

where E_C is the minimum of the conduction band and E_V is the maximum of the valence band. If this band gap is sufficiently small - typically less than 3 eV to 4 eV (varies in literature) - excitations can promote electrons across the gap, generating mobile charge carriers in both the valence band and the conduction band. This property defines the material as a semiconductor, and implies that its conductivity is sensitive to external factors such as temperature and external voltages.

Silicon is a semiconductor with a band gap of 1.12 eV. It has an indirect band gap, meaning the conduction band minimum and the valence band maximum occur at different

points in $\vec{k}$-space. Consequently, electron transitions across the band gap require both a photon of 1.12 eV and a phonon, or alternatively, a single photon with higher energy.

Charge carriers in semiconductors can either be intrinsic, generated by direct excitations across the band gap, or extrinsic, introduced by doping the material with other elements. Group IV semiconductors, such as silicon, can be doped with group III or V elements.

Group III elements have one fewer valence electron than group IV elements, which creates additional electronic states above the valence band. These states are empty at 0 K, but can be occupied by electrons from the valence band at temperature $T > 0$. The excitations leave behind unoccupied electronic states - equivalent to free holes - which act as positive charge carriers in the valence band. This is known as p-type doping, because the majority charge carriers (holes) carry positive charge. A small number of electrons may also be excited into the conduction band, acting as minority charge carriers.

Analogously, doping with group V elements is called n-type doping for the negatively-charged majority carriers (electrons). The additional valence electrons from dopants can occupy new states below the conduction band boundary, and can be excited to the conduction band when $T > 0$.

Both p-type and n-type doping introduce additional extrinsic charge carriers and improve the conductivity of semiconductors, particularly around room temperature. Furthermore, conductivity can be controlled by the adjusting the doping concentration, making this technique fundamental to modern semiconductor technology.

3.1.2 P-N Junction

A p-n junction is an interface between a p-doped and a n-doped semiconductor. It functions as a diode with distinctive electrical behaviours and serves as a fundamental building block in many electronic applications.

As discussed in the previous section, p-type and n-type semiconductors have different majority charge carriers. When the two are connected, gradients of charge carrier densities are established. The electron density is high on the n-side and low on the p-side, while the hole density shows the opposite trend. This causes electrons and holes to diffuse in opposite direction, towards each other and recombine near the junction. Each recombination removes a pair of electron and hole, and leave behind dopant ions: positive n-dopant (donor) ions and negative p-dopant (acceptor) ions. These immobile ions create an electric field that counteracts the carrier gradients and stops further diffusion. The region near the interface becomes depleted of free charge carriers - known as the depletion zone.

The effects of the carrier concentration gradient and the dopant ions in a p-n junction can be described with Poisson equation and Fick's law of diffusion. Solving these equations

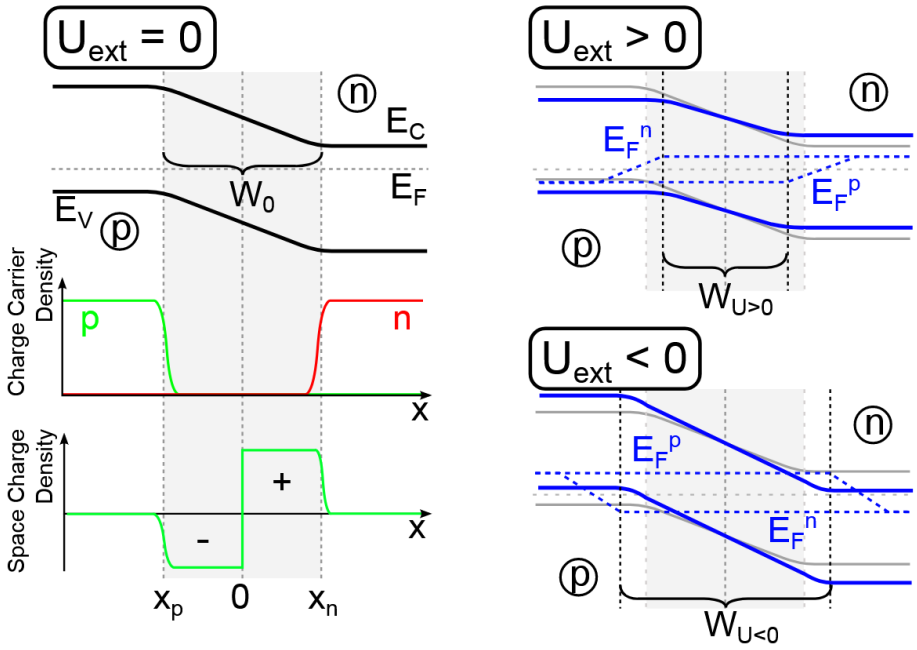


Fig. 3.1 Band diagrams of a p-n junction under different bias voltages[14]. The plots of charge carrier density and space charge density of a unbiased junction are also shown on the left. w is the width of the depletion zone, $x_{n,p}$ are the its widths in n,p substrates respectively. The charge carrier density is zero inside the depletion region, whereas the total charge density is zero outside. The width changes according to the sign and magnitude of bias voltage.

alongside the total charge density and boundary conditions, it's possible to determine the width of the depletion zone, w[1]:

$$w = (x_p + x_n) = \sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_D + N_A}{N_D N_A}V_{contact}}, \tag{3.2}$$

where $N_{A,D}$ are the dopant densities of acceptor (p) and donor (n) respectively, $V_{contact}$ is the contact potential[2], given by

$$V_{contact} = V(x_n) = \frac{2\varepsilon_0\varepsilon_r}{q}(N_D x_n^2 + N_A x_p^2) \tag{3.3}$$

where ε_r is the relative permittivity and q is the electric charge of the carriers.

[1]For a detailed derivation, see [15] or [16].

[2]Contact potential is the potential difference across a p-n junction at equilibrium and without external bias voltage. Also known as the built-in voltage.

The width of the depletion zone can be modified by applying external bias voltages, V_{ext}:

$$w = \sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_D + N_A}{N_D N_A}(V_{contact} - V_{ext})} \tag{3.4}$$

if the bias voltage is small comparing to the contact potential[15]. The width decreases for a forward bias ($V_{ext} > 0$) and increases for a reverse bias ($V_{ext} < 0$).

3.1.3 Application in Particle Sensing

When a particle traverses a semiconductor, it generates charge carriers along its path, through processes such as electron-hole pair production and scattering. These charges can be efficiently separated if they are generated in the depletion region, where a non-zero electric field drives the electrons and holes in opposite directions, towards the collection electrodes.

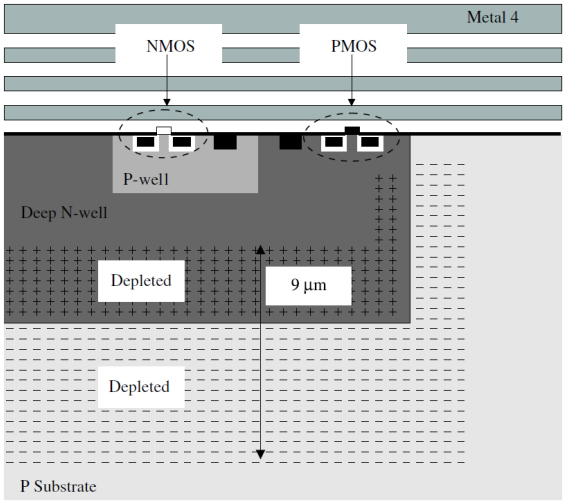
In particle sensing applications, a wider depletion region increases the effective path length of traversing particle in the p-n junction, thereby increase the charge deposit and improve detection efficiency. This is achieved by applying reversed bias, which expands the depletion zone. However, if bias is too high, the resulting E field will be strong enough to ionise charge carriers, generating additional electron–hole pairs. This leads to an avalanche effect, causing breakdown of the sensor.

3.2 HV-MAPS-based Detectors

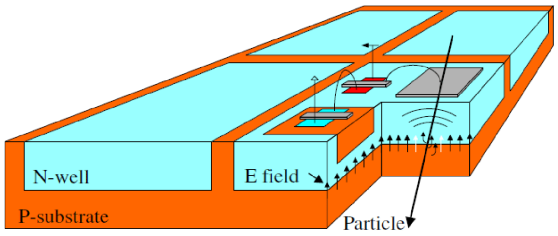
HV-MAPS are fabricated using commercial CMOS technology. "High-voltage" refers to the bias voltage between the deep n-well and the substrate, typically exceeding 50 V. "Monolithic" indicates that the sensors are integrated, with the amplifier, comparator, and readout electronics built directly on the same silicon die as the sensing elements. Standard low-voltage CMOS transistors are used for the internal electronics of the chip, and floating logic is employed to bridge the high-voltage sensor with the low-voltage CMOS circuit. HV-MAPS offer several advantages, including radiation hardness, lower costs, and compact functionality, making them well-suited for particle detection and tracking applications[13].

The floating logic structure, illustrated in Fig. 3.2, is used to implement the pixel architecture in HV-MAPS. The pixel's sensing element is a p-n junction formed by a deep n-well within a p-doped substrate. It is reverse-biased with a high voltage close to the breakdown voltage, to extend the depletion region deep into the lowly doped substrate. For example, under a 60 V bias, the depletion zone can reach a thickness of approximately 9μm. A particle traversing this region can generate relatively large signal, on the order of 700

e[3]. The high electric field in the depletion zone allows charge carriers to be collected by drift, with the n-well serving as the cathode. This drift-based collection enables faster signal response comparing to low-voltage triple-well MAPS, where charge collection relies on slower diffusion mechanisms.



(a) The floating logic, including the sensing diode. The plus and minus signs denote the area with non-zero total charge density - the depletion region.



(b) Four high-voltage pixels.

Fig. 3.2 The pixel structure of HV-MAPS[3].

An arbitrarily complex CMOS readout circuits can be implemented within the deep n-well, allowing fully integrated signal processing. A simplified block diagram of the pixel architecture is shown in Fig. 3.3. The separated charges are first collected by the p-type and n-type MOSFET (PMOS and NMOS) embedded within the N-well, producing a short current pulse. During this stage, charge sharing between neighbouring pixels may occur. The signal is then processed by a charge-sensitive amplifier (CSA), which integrates the incoming charge and converts it into a proportional voltage signal. Next, the voltage is sent to a comparator, which compares the signal with the threshold (THL) value and digitises it. Finally, the signal is passed to the readout, in which pixel address, time-of-arrival (ToA), and collected charge are assigned.

The charge of a signal is represented by threshold-over-time (ToT), which is linearly proportional to its energy and can be calibrated at a test beam. The readout is triggered when the output voltage exceeds the predefined threshold. The rise time of a signal - the time taken to reach the peak output voltage - depends on factors such as the capacitance and transconductance of the amplifier, but is independent of the input charge (see [17]). The amount of charge affects only the peak output voltage. As a result, high-charge signals

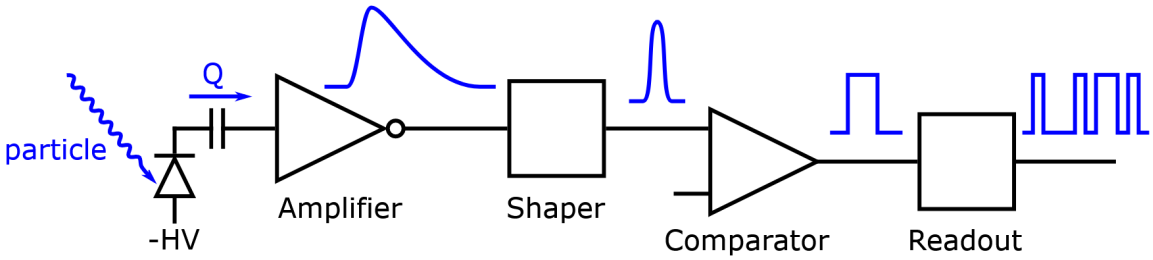
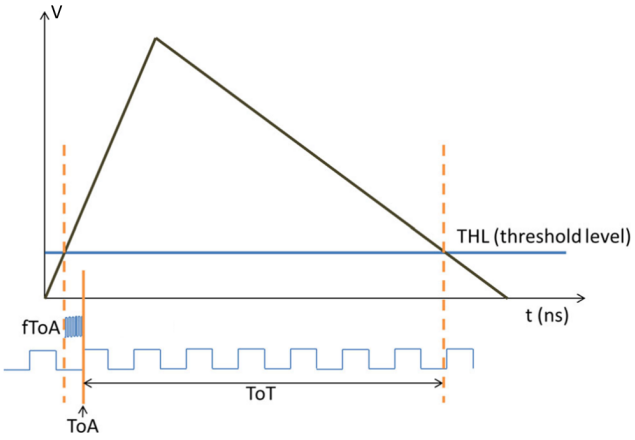


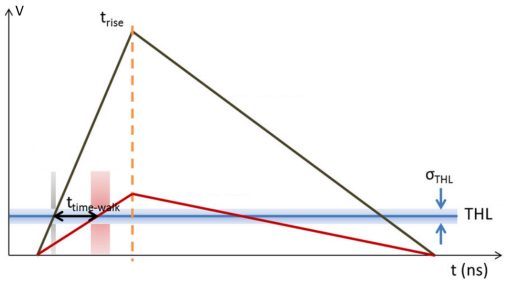
Fig. 3.3 A simplified block diagram of the pixel[16]. The signal processing and readout circuitry are integrated in the same silicon die as the sensor diode.

remain above the threshold for longer duration than those with lower charge, as shown in Fig. 5.5. Therefore, ToT is a reliable proxy for the deposited charge.

However, for the same reason, the time at which a signal crosses the threshold also depends on its amplitude. Since this crossing time is used to determine signal's time-of-arrival (ToA), the measured timing becomes dependent on ToT. Low-charge signals exhibit larger time delays comparing to high-charge signals. This phenomenon is known as the time-walk effect, and must be corrected for to achieve accurate timing.



(a) Timing measurement of a single signal, showing the threshold crossing used to determine the Time-of-Arrival (ToA).



(b) An illustration of the time-walk effect. The brown line represents a high-charge signal, while the red line represents a low-charge signal. The difference in threshold-crossing times demonstrates the charge-dependent delay in signal detection.

Fig. 3.4 Illustrations of timing characteristics in HV-MAPS readout[18].

3.3 MightyPix and TelePix2

Building on the developments of MuPix[19][20] and ATLASPix[5][6], MightyPix is a family of HV-CMOS MAPS sensors designed to meet the requirements of radiation hardness, timing,

and granularity in the central pixelated region of the Mighty Tracker Stations. MightyPix1 is the first HV-MAPS sensor designed for LHCb[21], developed by the ASIC and Detector Laboratory at the Karlsruhe Institute of Technology (KIT)[22].

The performance of MightyPix is interested; however, only 16 working chips are available, which is insufficient to carry out a radiation damage study. Instead, TelePix, a related family of HV-CMOS MAPS with a similar design[3], is more widely available. In this project, the time resolution of TelePix2 was evaluated to gain insight into the timing performance of sensors of this type.

The TelePix family is designed to serve as telescope planes in the DESY II Test Beam Facility to solve the problem from separated telescope and trigger plane[9]. It is designed by KIT and further developed in collaboration with Heidelberg University and DESY[24]. The latest version, TelePix2, offers several key features, including high-resolution time sampling, fast signal output, a user-configurable trigger (e.g., pixel masking), and low material budget. The parameters are summarised in Table 3.1.

Technology	Pixel dimension	Chip dimension	Pixels per chip
180nm HV-CMOS	$165 \times 25\,\mu m^2$	$20 \times 13\,mm^2$	120×400

TDC bandwidth	Maximum HV bias	Material budget
125Hz(4ns)	-90V/-150V (pixel/chip)	0.0011 from $100\,\mu m$ silicon

Table 3.1 The parameters of TelePix2[9][25].

[3]For a comparison between MightyPix and TelePix2, please refer to [8][23].

Chapter 4

Event Reconstruction from Test Beam Data

This chapter describes the test beam setup for data collection in this project, and outlines the procedures for extracting key physical quantities from the raw data - event reconstruction.

4.1 Test Beam Setup

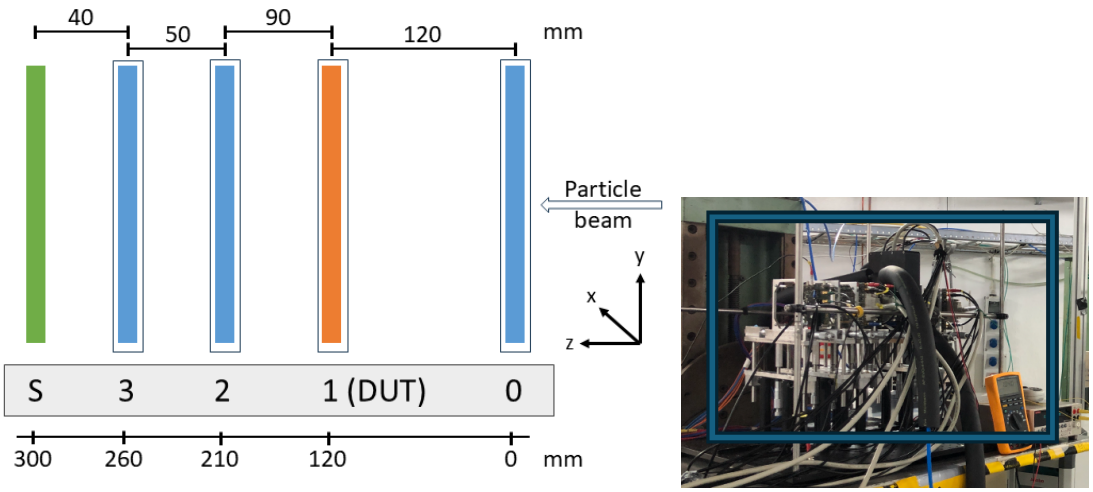
In general, the time resolution of a sensor is determined by the width of the time residual distribution, which equals the difference between the arrival time of a particle at the sensor, t_{dut}, and a reference arrival time, t_{ref}:

$$\Delta t_{residual} = t_{ref} - t_{dut} \tag{4.1}$$

A test beam provides high-energy particles capable of traversing multiple sensor layers, allowing spatial and timing information to be measured across the setup. This enables track reconstruction and time comparison, making it essential for time resolution analysis.

The test beam measurement was performed at the DESY II, Germany, in March 2024 by Tianqi Gao et al[26]. The setup consists of four sensor planes and one scintillator, as shown in Fig. 4.1. An electron beam of energy 4 GeV was passed through the setup.

A MuPix11 telescope with MuDAQ readout was used. MuPix11 is a CMOS HV-MAPS that was designed for the Mu3e experiment, with a time resolution of 14.3 ns[19], a spacial resolution of 23.1μm$\times$23.1μm and an active area of 256$\times$250 quadratic pixels of 80μm pitch[20]. Similar to TelePix2, MuPix11 is capable of recording 2D spacial coordinates, ToA and ToT of the incoming signals. There are three MuPix11 planes in the telescope, where the



(a) The telescope.

(b) A picture of the setup[26].

Fig. 4.1 Schematic of the test beam setup. The orange layer represents the Device under Test (DUT), which is the TelePix2 sensor. Blue layers are the telescope layers, and the green layer is the scintillator. The rightmost (0th) layer serves as the reference layer.

first plane upstream serves as the reference plane. They provide reference spatial information for track reconstruction.

The Device under Test (DUT) - TelePix2 - was placed between the first and second MuPix11 plane. The scintillator was placed downstream and has a time resolution of approximately 1 ns. It provides ToA of the signal, because the time resolution of TelePix2 was expected to be better than that of the telescope.

4.2 Event Reconstruction with Corryvreckan

A typical run in the experiment contains approximately 10^6 hits on each of the five layers. To obtain a distribution of time residuals, it is essential to identify the path individual particles take through the setup. This is achieved through a process called event reconstruction, where coincidence events from different detector layers of the telescope are used to reconstruct the particle trajectory. It involves identifying spatially and temporally consistent hits across the layers, and grouping them into tracks. For test beams, the particles are usually highly relativistic, thus scattering is negligible and the tracks are taken to be straight lines. After a complete reconstruction, statistics can be gathered on the ToA difference between the DUT and the scintillator, by analysing a large number of individual particle tracks.

For the purpose of this project, the Corryvreckan framework[10] was used for reconstruction. Corryvreckan is a 4D reconstruction and analysis software based on modern C++,

initially developed by the CLICdp collaboration at CERN[27]. It is based on a modular design, where each step of the reconstruction process is performed separately by user-configurable modules. A flow chart of a typical Corryvreckan reconstruction chain is shown in Fig. 4.2. Each step will be briefly explained in the following. More detailed discussions on the architecture of the framework can be found in [28][29][30].

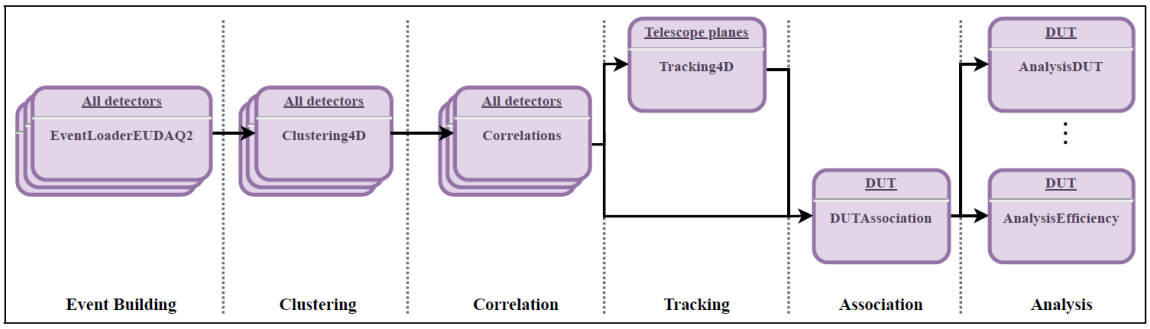


Fig. 4.2 A typical Corryvreckan reconstruction process[28]. Each box represent a step in the reconstruction, which contains the corresponding Corryvreckan module. The underlined title refers to the sensor planes that the module applies to.

4.2.1 Event Building

The raw data files are coded in binary form. At the start of a reconstruction, decoding is needed for them to be processed by subsequent modules. This is performed by a group of modules called `EventLoader`. Here, for the MUDAQ readout, the corresponding `EventLoaderMupixTelescope` module was used.

Next, the data is divided into time slices know as "events" to establish a unified time structure for all sensors. The event definition can vary depending on the purpose of the analysis[1]. In this project, the `Metronome` module was used to define successive events of equal length 16μs, as illustrated in Fig. 4.3. All subsequent modules obey this event definition, and provide only matching data[2].

4.2.2 Clustering

A cluster is a collection of adjacent pixel hits that originate from a single traversing particle, due to charge sharing. Clustering is the process of identifying and grouping the hits into

[1]For a more detailed description of event definition based on individual detectors, see [28].

[2]This is because the hit information are loaded onto the central clipboard storage on an event-basis, and the subsequent modules can only read information from the clipboard. In other words, the reconstruction steps are carried out within an event.

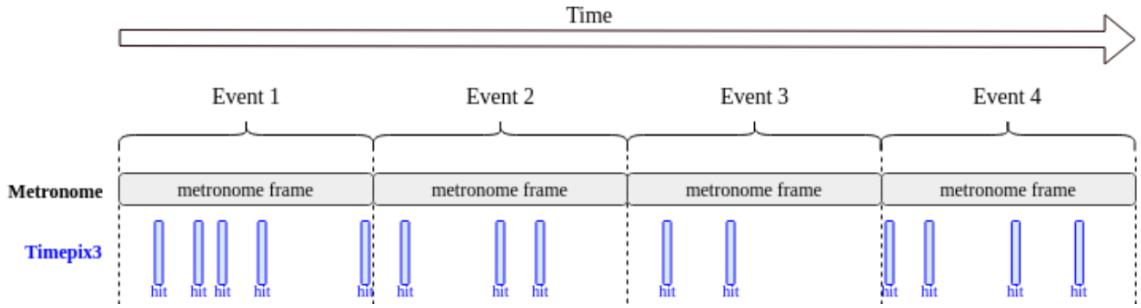


Fig. 4.3 The event building mechanism based on `Metronome`[30]. It defines the events, and the subsequent `EventLoaderTimepix3` module loads all signals with suitable timestamp to the central clipboard storage.

clusters. The `Clustering4D` module performs a nearest-neighbour search for all pixel hits on a sensor plane, and produces clusters accordingly. This is performed for all sensor planes. The cluster position is determined by taking the arithmetic mean of pixel hit positions within the cluster. The cluster timestamp is taken to be the timestamp of the highest-charge pixel. The charge of the cluster is the sum of pixel charges.

4.2.3 Track fitting

There are two steps in the track fitting process: the first is to find all possible track candidates using one of the `Tracking` modules; The second is to perform fitting with the track object, where bad fits are discarded and the result is reported back to the module.

The `Tracking4D` module was chosen for the process. For every signal detected on the first (upstream) sensor plane and each signal on the last plane, a straight line is connected between them - which is a primary track candidate. It is then possible to determine where and when the track candidate intercepts with the remaining planes.

Next, the module searches for clusters on these remaining telescope planes that are within a configurable spatial and time window[3]. In this analysis, the windows were set to be $500\mu\text{m}\times500\mu\text{m}$ and 200 ns for good statistics. Only when matching clusters are found on all telescope planes, a track can be assigned to these signals.

Finally, the remaining tracks are added to the clipboard. Each track timestamp is calculated by the average timestamp from all clusters in the track.

[3]The time window should be dependent of the distances between layers. However, the Time of Flight is usually in the order of 10^9s and negligible comparing to time cut.

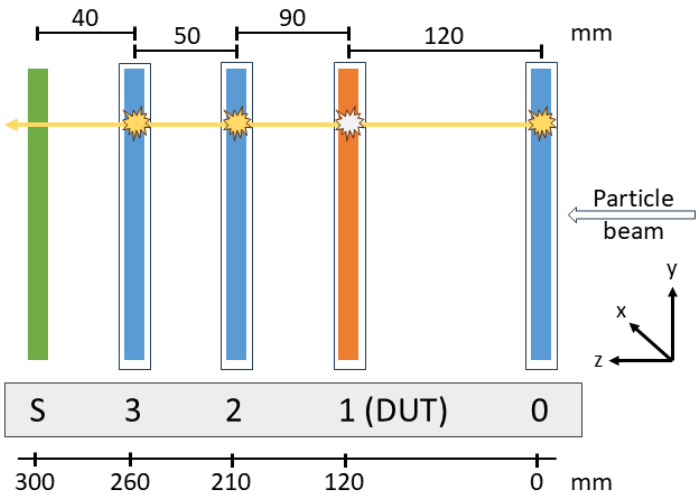


Fig. 4.4 Schematic of fitting of a straight track. The track is marked as an yellow arrow. The yellow clusters are those responsible for fitting the track. The white cluster is the DUT cluster that is associated to this track.

4.2.4 Association of DUT Clusters with Tracks

After track constructions, the `DUTAssociation` module searches for clusters on the DUT around the intersection point of each track. Analogous to `Tracking4D`, clusters within the spatial and temporal cut, which was chosen to be 500μm$\times 500\mu$m and 500 ns respectively, will be associated to the track.

The efficiency of the DUT can be calculated from these tracks:

$$\text{Efficiency} = \frac{\#\text{tracks with associated clusters on DUT}}{\#\text{total tracks through DUT}}. \tag{4.2}$$

4.2.5 Analysis of the DUT

Before the final analysis, corrections to the setup geometry are required, as it was measured manually at the test beam. A detailed description of the alignment procedure is provided in Appendix C. Once the telescope and DUT have been successfully aligned, the reconstructed data can be considered reliable and used for subsequent analysis.

The `AnalysisEfficiency` and `AnalysisDUT` modules perform analysis and calculate key quantities about the performance of the DUT, such as efficiency and time resolution. Most Corryvreckan modules generate ROOT histograms[31], from which analysis plots can be directly accessed.

4.2.6 Correction for Time-walk Effect

In this project, correction for the time-walk effect is calculated by a stand-alone python file developed by Mary Richardson-Slipper et al[32]. For each Time-over-Threshold(ToT) value, a vertical slice is taken from the 2D histogram of time residual versus ToA. The width of the slice was set to be 20 ns, to balance fitting accuracy and statistics.

Each slice is then projected on to the time residual (y) axis, giving a 1D histogram of time residual, as shown in Fig. 4.5. A crystal ball function is fitted to the distribution, and the mean of the fit is taken as the correction value for that ToT. To validate the correction, the fitted values are compared against an empirical model for the time-walk effect[18]:

$$\Delta t_{timewalk} = \frac{a}{(E+b)^d} + c \tag{4.3}$$

where E denotes the ToT, and a, b, c, d are fit parameters.

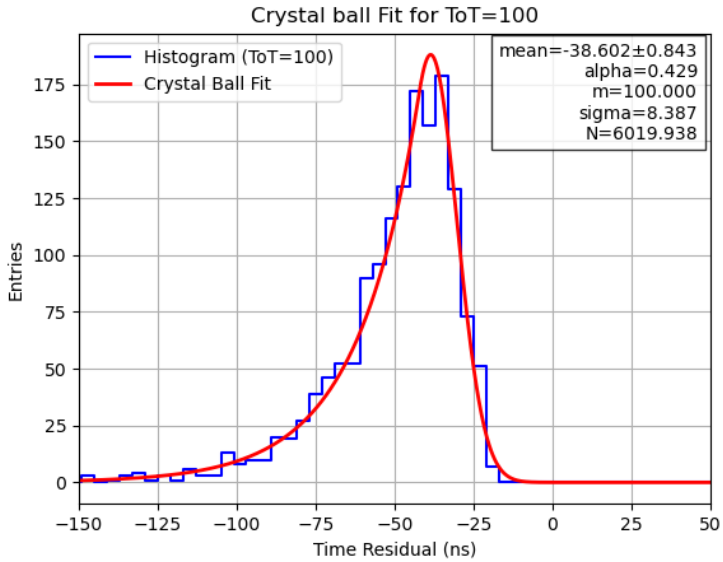


Fig. 4.5 An example crystal ball fit, for a slice of width 20 ns at ToT=100 ns, from Fig. 4.6. The correction for this ToT value is taken to be the fitted mean value, -38.602 ns.

It is worth noting that at high ToT values, the time residuals are not centred at zero. This offset arises from a constant time difference between the DUT and scintillator timestamps. The correction method described here have accounted for both this offset and the time-walk effect.

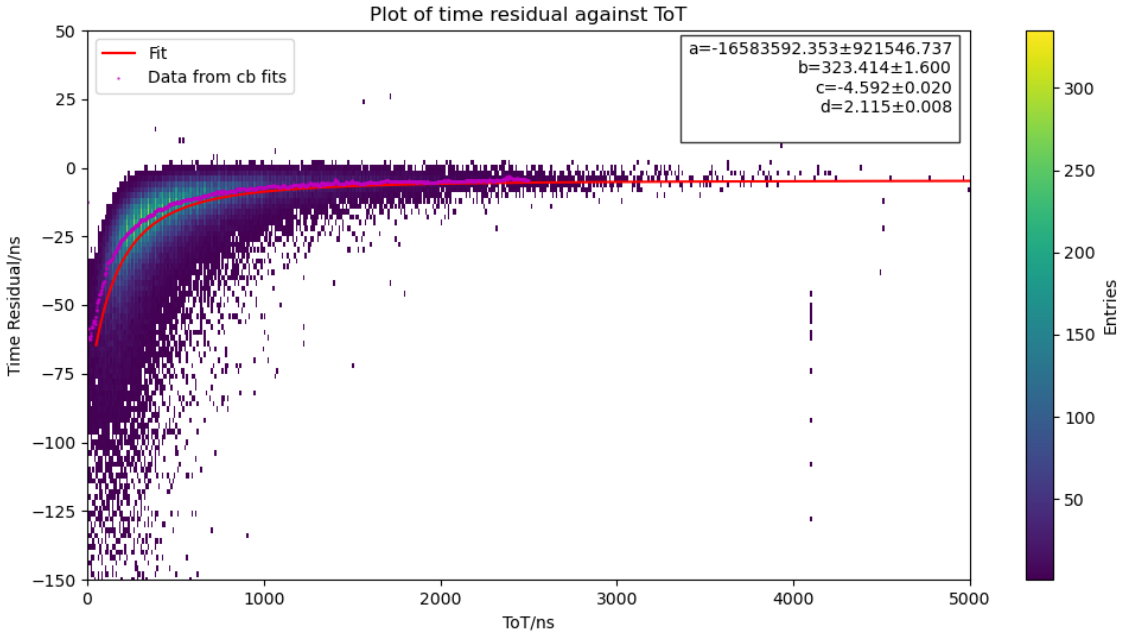


Fig. 4.6 An example plot of time residual versus Time-over-Threshold(ToT), where the colour map represent the entries at each point. The curvature at low ToT values illustrates the time-walk effect. Purple markers refers to the correction values calculated from crystal ball fits, while the red curve is a fit to Eq. 4.3. Both methods yield a consistent trend with slight deviations at low ToT values. Note that the threshold in this run was 200 DAC, which is significantly higher than typical values used in data-taking - purely to emphasis the time-walk effect.

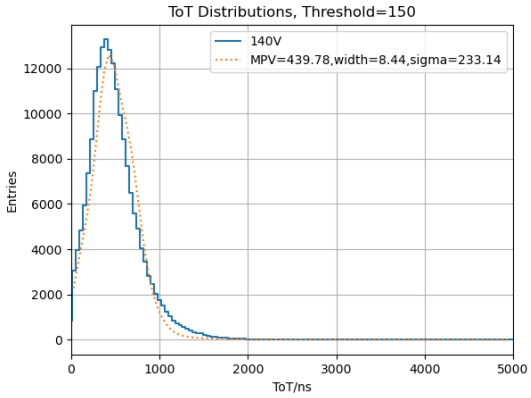
Chapter 5

Results and Discussions

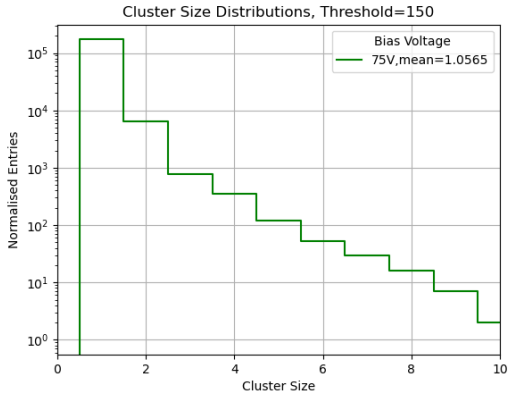
5.1 ToT and Cluster Size Distributions

Runs with varying pixel bias voltages and readout thresholds were analysed to obtain the distributions of time-over-threshold (ToT) and cluster size. This step is a quality assurance check, verifying that the observed trends with respect to changes in bias and threshold voltages are consistent with expectations.

Each ToT distribution was fitted with a Landau–Gaussian convolution to extract the peak value, representing the most probable charge deposition. For the cluster size distribution, the mean cluster size was calculated as a measure of the overall distribution, and charge sharing effect may be estimated. As shown in Fig. 5.1, the distribution was dominated by single-pixel clusters. While charge sharing was evident, the entries of larger clusters decreased rapidly with increasing size.



(a) The distribution of ToT.



(b) The distribution of cluster sizes.

Fig. 5.1 The distributions from a run with bias voltage 75V and threshold 150 DAC.

Next, the peak ToT values were compared cross different bias voltages and thresholds in Fig. 5.2. In general, the peak value increases with higher bias voltage, due to increased charge production in a wider depletion zone, as expected. Conversely, the peak ToT decreases with higher threshold: for a given signal, a higher threshold reduces the time the signal remains above threshold. The outliers may result from unstable fits of the convoluted Landau-Gaussian function, which were not further refined due to time constraints.

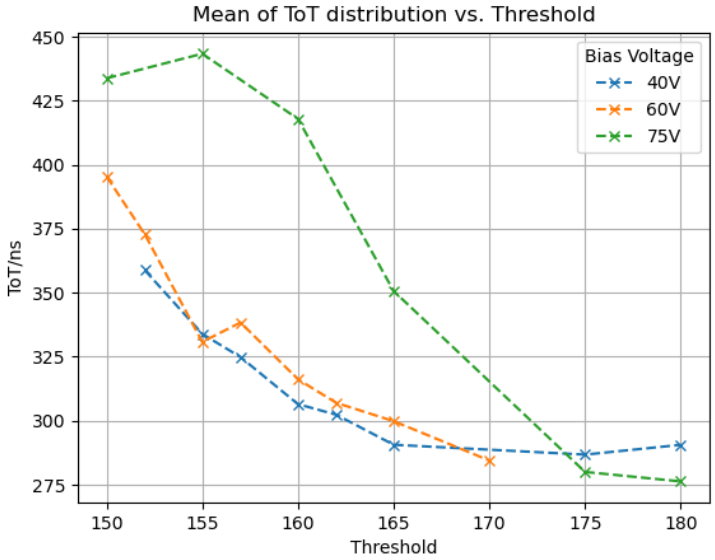


Fig. 5.2 A plot of the mean ToT value of the fitted Landau distribution versus the threshold.

A similar comparison was performed for the mean cluster size in Fig. 5.3. A minimum appears around a threshold of approximately 155–165 DAC, consistently observed across all bias voltages, with the position of the minimum shifting horizontally. A possible explanation is that, at these threshold values, noisy pixels near a real cluster fall below the threshold and are therefore not registered. When the threshold is further increased, single-pixel clusters begin to be suppressed, leaving only clusters with very large charge deposits and size. This leads to a rise in the mean beyond the minimum and a reduction to the number of events, as indicated by the larger error bars. This observation has also been reported in a separated experiment of TelePix2[33].

Overall, the performance of TelePix2 was stable and largely consistent with expectations, though improvements to the fittings are suggested.

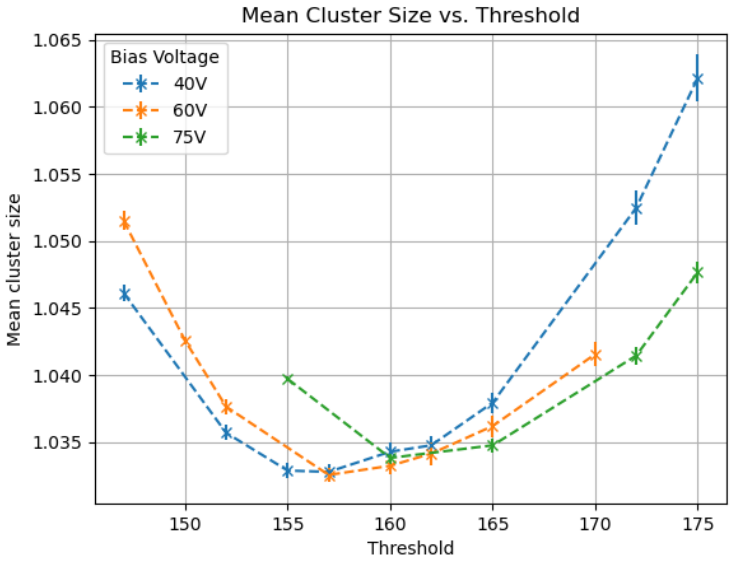


Fig. 5.3 A plot of the mean cluster size versus the threshold.

5.2 Efficiency

To determine the optimal combination of bias voltage and threshold that gives the most reliable data for timing measurements, the efficiency defined in Eq. 4.2 was calculated.

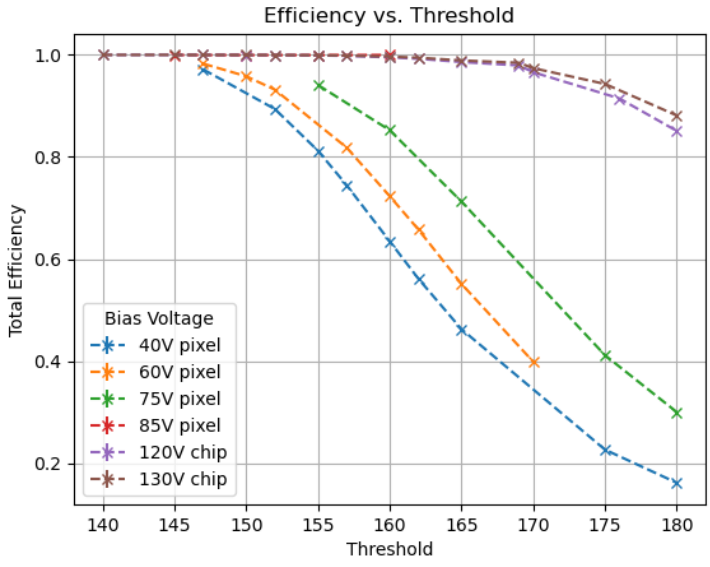


Fig. 5.4 The plot of efficiency versus threshold, at different bias voltages: 40-85 V pixel bias (on irradiated sensors), 120-130 V chip bias (on not-irradiated sensors). The difference between pixel and chip bias is given in Appendix D.

The sensor demonstrated good performance (efficiency≥96%) for threshold voltages below 160 DAC and bias voltages≥85 V. Based on these observations, the run with a bias voltage 130 V, threshold 140 DAC was chosen to calculate the time resolution.

Additionally, the results for pixel bias were obtained with sensors irradiated to a fluence of $1 \times 10^{14} n_{eq}$, demonstrating good radiation tolerance. It also suggests that a pixel bias above 85 V is required to ensure reliable operation, as the sensor accumulates radiation damage over its lifetime.

5.3 Time-walk Correction

As described in Sec. 4.2.6, the time-walk effect requires correction in low ToT regions where it is significant. However, within the relevant range in the chosen run, the statistics was very limited - approximately 1% of the entries fall within this region. This indicates that with an appropriate threshold, time-walk correction is not necessary for TelePix2 test beam.

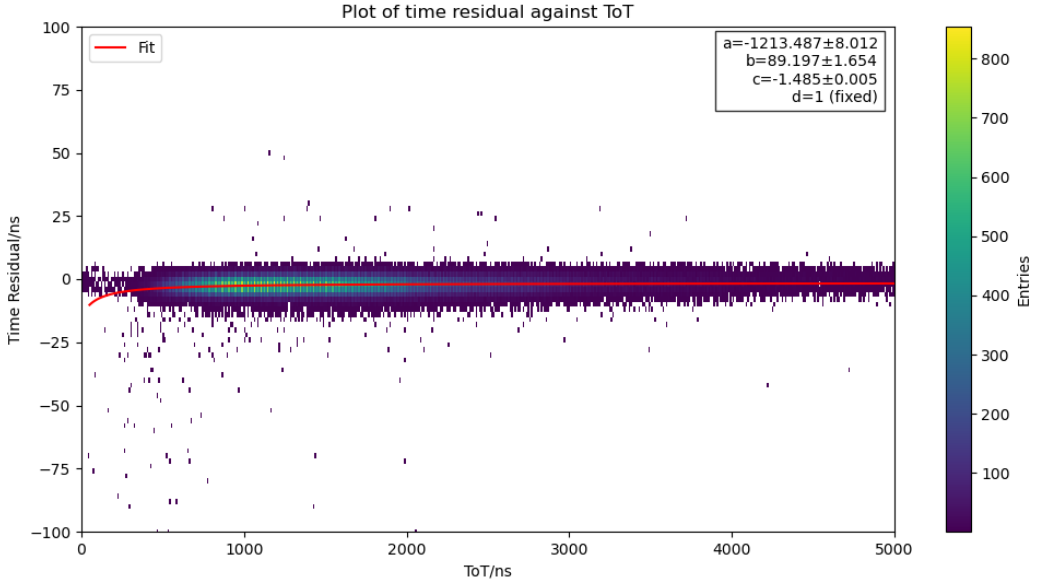


Fig. 5.5 The 2D histogram of time residual versus Time-over-Threshold(ToT).

It is worth noting that the time-walk effect becomes increasingly pronounced at higher thresholds. For a fixed bias voltage, the effect becomes noticeable from approximately 150 DAC onwards.

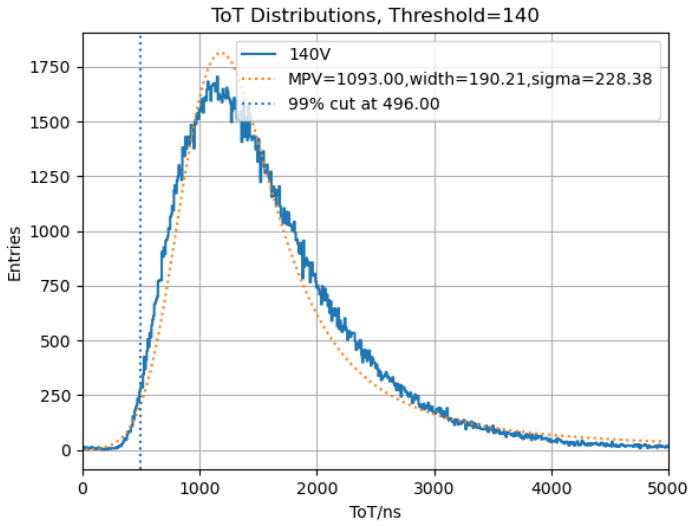


Fig. 5.6 The plot of efficiency versus threshold, at different values of bias voltage.

5.4 Time Resolution

A Gaussian curve was fitted to the time residual distribution for the chosen run. The time resolution was measured to be 2.656 ± 0.003ns, which satisfied the requirement of <3 ns.

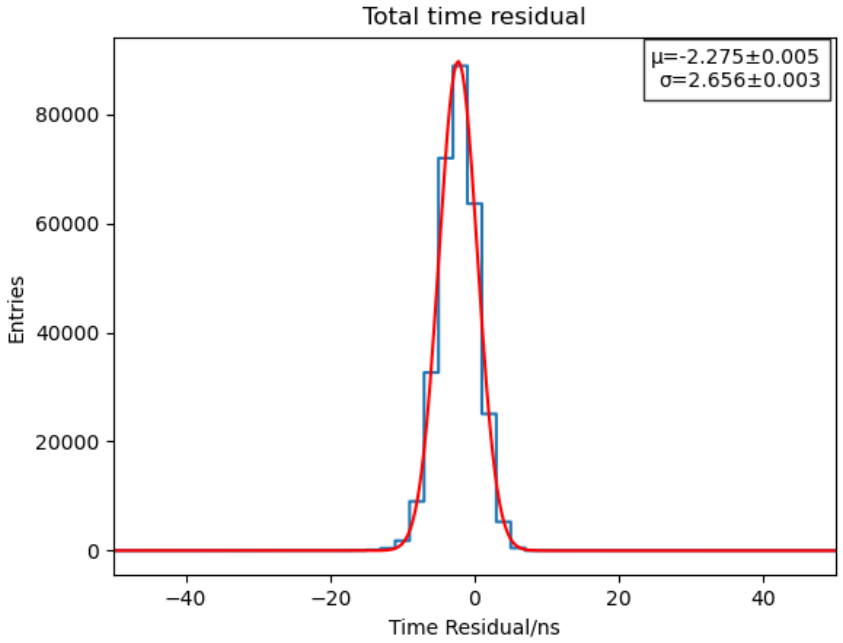


Fig. 5.7 Time resolution of TelePix2, with chip bias voltage 130V, threshold 140 DAC.

5.5 In-pixel Spatial Correlation

Finally, the entries of 1-pixel and 2-pixel clusters were plotted as a function of the in-pixel x position to investigate the charge-sharing effect. It is clear in Fig. 5.8 that the frequency of 1-pixel clusters decreases towards pixel edges, where charge-sharing is more likely to occur. However, the distribution of 2-pixel clusters has low statistics and does not exhibit a clear positional trend, suggesting the need for a longer data taking.

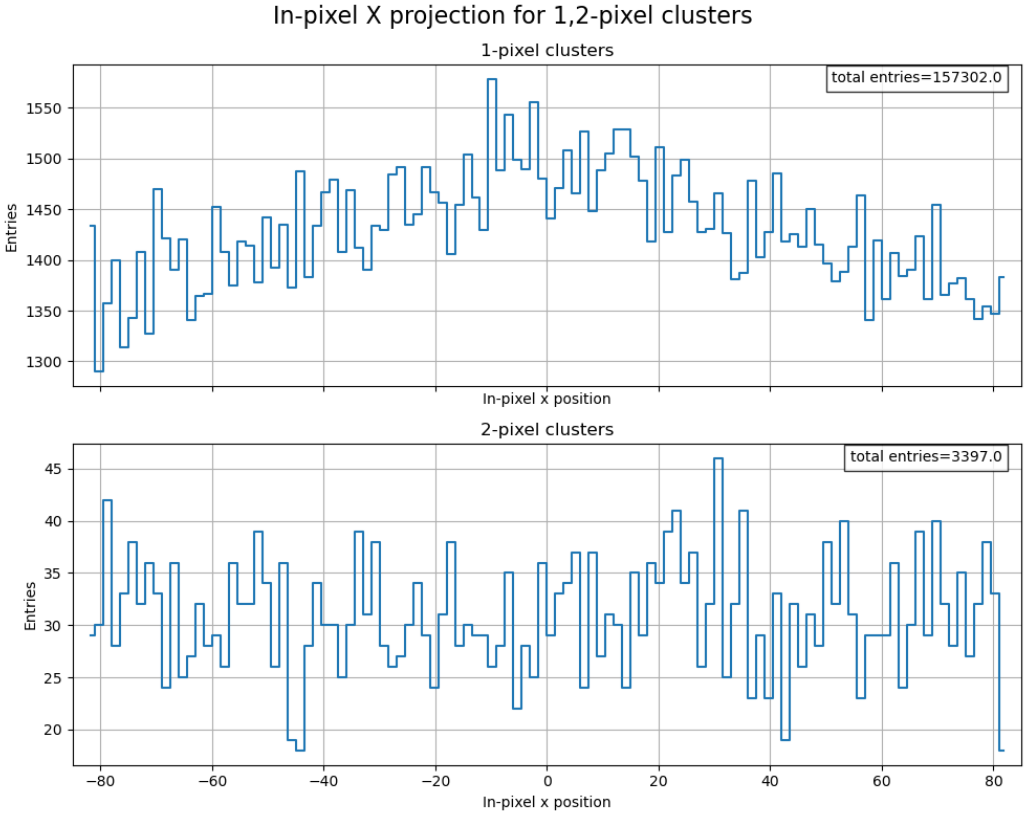


Fig. 5.8 A plot of the cluster entries versus their x-positions inside a pixel, for 1-pixel and 2-pixel clusters.

Chapter 6

Conclusion

A detailed analysis was conducted on the time-over-threshold (ToT) and cluster size distributions across runs with thresholds ranging from 140-180 DAC and bias voltages 40-85 V (pixel), 120-130 V (chip). The ToT distributions were fitted with Landau–Gaussian convolutions to extract their peak values, which were found to increase with bias voltage and decrease with higher thresholds, in line with expectations. Mean cluster sizes exhibited a minimum around a threshold of 155 DAC, with the position of this minimum shifting toward higher values as the bias voltage increased. This is due to noise clustered to real signals at lower thresholds, and real clusters being cut away at higher thresholds.

The efficiency of TelePix2 was also evaluated, suggesting high efficiency ($\geq$96%) with bias $\geq$ 85 V and threshold 140-160 DAC. The configuration with 130 V chip bias and 140 DAC threshold was selected for the time resolution measurement. Under these conditions, the impact of time-walk effect was small - only around 1% of the data points fall below a ToT value of 500 ns, and the magnitude of the shift is negligible. The resulting time resolution was measured to be 2.656 ± 0.003 ns, meeting the 3 ns requirement. However, time-walk effects become more pronounced at thresholds above 150 DAC, in which case corrections would be necessary.

Future studies can proceed along several directions. On the experimental side, a wider scan of bias voltages - beyond the limited range available in this study - should be conducted. Additionally, performing test beam experiments with X-ray sources could enable calibration of ToT with respect to deposited energy.

On the analysis side, it would be valuable to investigate how time resolution depends on both bias voltage and threshold. Time-walk corrections should be applied to datasets where the effect becomes non-negligible. These analyses can be carried out using existing data and scripts. In addition, the quality of the Landau-Gaussian fits can be improved by further optimising the initial fitting parameters.

In the longer term, when more MightyPix sensors become available, direct testing under conditions representative of LHCb environment should be undertaken to evaluate their performance.

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Appendix A

Structure of Code

The code used in this project is based on the repository by Mary Richardson-Slipper et al[32]. A link to the repository can be found here. Required packages can be installed by following the instruction in the `README` file. For beginners, a more detailed guide, including solutions to some frequently-asked questions, is available here.

The structure of the repository is as follows:

Name of folder	Description
`scripts`	Linux bash scripts for event reconstruction and Python scripts for time-walk correction and further analysis.
`output`	Output ROOT files of Corryvreckan without time-walk correction: ROOT files with histogram plots.
`output/plots`	Plots produced by Python scripts.
`output/timewalk_correction`	ROOT files with `TGraph` objects that records the time-walk values calculated by crystal ball fits, including error bars.
`output/scans`	Output ROOT files of Corryvreckan with time-walk correction.
`config/for_scripts`	Configuration files for Corryvreckan modules.
`config/geo_prealign`	Geometry files, need to be updated manually after prealignment.
`config/geo_results`	Geometry files, which are updated automatically by running Corryvreckan.

The Linux bash scripts involved in event reconstruction are:

Name of script	Description and instruction after running the sciprt
`single_prealign`	Prealignment. Check the spatial correlation plots in the output ROOT file, then update the geometry file with correlations of the telescope planes.
`align_telescope`	Telescope alignment. Check the χ^2 and global spatial residuals plots of the telescope planes in the output ROOT file. If not ideal, update the geometry file and run again.
`align_DUT`	DUT alignment. Check the spatial correlation plots of the dut in the output ROOT file, then update the corresponding geometry file and run again.
`check_DUT`	Analysis of DUT, only run when the alignments are complete. The data may be used for further analysis if time-walk correction is not required.
`timewalk_correction`	Correction for time-walk effect. Check if the correction is sensible, then the data in the output ROOT file may be used for further analysis.

The python scripts for further analysis and plotting are:

Name of script	Description
`check_tot.py`	Fit a Landau-Gaussian convolution to a chosen ToT distribution. Find the ToT values where 99% of total entries lies above.
`compare_tot.py`	Plot ToT distribution for different pixel bias voltages at a given threshold.
`compare_totMPV.py`	Plot the peak position of ToT distribution against threshold, at different bias voltages.
`compare_cluster_size.py`	Plot cluster size distributions for different pixel bias voltages at a given threshold.
`compare_cluster_size_mean.py`	Plot mean cluster size against threshold, at different bias voltages.
`get_residual_vs_tot.py`	Plot a 2D histogram of time residual versus ToT, entries are represented by colour map.
`get_time_resolution.py`	Plot the time residual distribution, fit a Gaussian curve to find μ and σ.
`get_timewalk_effect.py`	Demonstrates the effect of applying time-walk correction to a slice of the time-residual-vs-ToT plot.
`get_inpixel.py`	Plot the entries of clusters of a given size versus the in-pixel x position.
`compare_effciency.py`	Plot efficiency against threshold, at different bias voltages.

Appendix B

MOSFET

MOSFETs (Metal-Oxide-Semiconductor Field-Effective Transistors) are the fundamental components of HV-MAPS technology. They apply the properties of p-n junctions to control the conductivity of a conduction channel between source and drain, through a gate-controlled external voltage.

The structure of a basic n-channel MOSFET (NMOS) is shown in Fig. B.1. "N-channel" refers to the electron-conducting channel, which will be discussed in detailed later. There are 4 connections for a MOSFET: the source, drain, gate, and substrate. In a NMOS, the substrate is typically p-doped, while the source and drain are n-doped, creating two p-n junctions: between source-substrate and between substrate-drain. These junctions are oriented such that current cannot flow direcly from drain to source. In contrast, for a p-channel MOSFET (PMOS), the substrate is n-doped and the source & drain are p-doped.

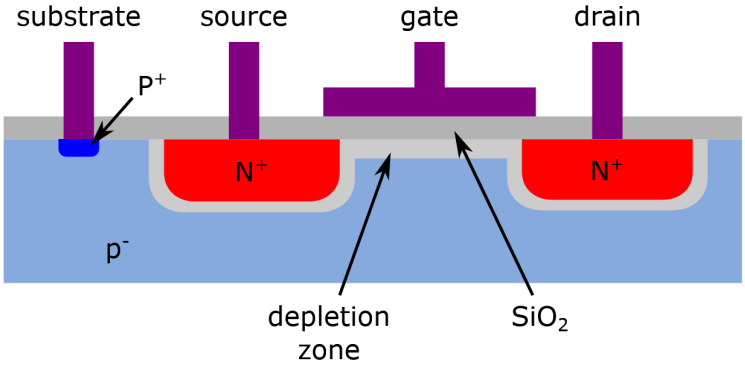


Fig. B.1 The sign on the doped regions refers to the concentration of doping[16].

The gate electrode is separated from the substrate by an insulating layer of silicon dioxide or ploy-silicon.

When no voltage is applied to the gate, the source-drain current is negligible. The MOSFET is considered "off".

When a positive gate voltage is applied (to NMOS), electrons in the p-doped substrate are attracted towards the interface between the substrate and the insulating layer. As the charge accumulates, a conduction channel begins to form between n-doped source and drain. At equilibrium, the the voltage of the accumulated carriers is equivalent to the gate voltage V_{gs}. Once V_{gs} exceeds the threshold voltage V_{th} - the voltage at which the number of free electrons equals to the number of holes in intrinsic silicon - an electron channel forms and source & drain get electrically connected. The MOSFET is considered "on". Similar arguments with opposite doping applies to p-channel MOSFET (PMOS)[1].

When the voltage between the source and the drain, V_{ds}, satisfies $V_{ds} < V_{gs} - V_{th}$, the MOSFET operates in the linear region. The conduction channel provides an ohmic connection, with current proportional to the voltage: $I_{ds} \propto V_{ds}$. When $V_{ds} = V_{gs} - V_{th}$, the channel pinched off at the drain. For $V_{ds} > V_{gs} - V_{th}$, the pinch-off point shifts towards the source, shortening the channel to compensate for the larger voltage. This marks the onset of the saturation region, where further increase in V_{ds} have little effect on I_{ds}.

In summary, the conductivity between the source and drain of a MOSFET is controlled by gate voltage. The device acts as a voltage-controlled resistor in the linear region, and as voltage-controlled current source in saturation region.

[1] For a PMOS, the substrate is n-doped and the minority carriers are the holes. When a negative gate voltage is applied, a hole-channel will be formed between the source and drain.

Appendix C

Alignment of the Setup

The configuration of the telescope was measured manually in the laboratory. For an ideal setup, sensor layers are aligned precisely along the beam axis, with positions of the form (0,0,z). However, manual assembly could introduce misalignments, particularly in the transverse (x,y) directions and rotations about the z axis. Displacements on the order of of 10^{-4}m are common, which are comparable to the pixel pitch of the sensor and could lead to incorrect or failed fits. If the misalignment reaches 10^{-3}m, track constructions may no longer be feasible. Therefore, the setup must be aligned by software before analysis can proceed.

The alignment procedure in Corryvreckan involves three steps, beginning with prealignment, which gives an initial estimate of the misalignment - typically on the order of 10^{-3}m to 10^{-4}m. This is done by calculating the spatial correlations, $x_{correlation}$ and $y_{correlation}$, where

$$x_{correlation} = x_{detector} - x_{reference} \tag{C.1}$$

refers to the local spatial difference[1] between clusters on the reference detector and those on target detectors. This is performed with the `Correlations` module. In an ideal setup, the peak of the correlation distribution would be centred around zero; in practice, there are typically offsets due to misalignment. However, for large initial offsets, an effective initial guess is to shift detector positions to align the correlation peak with zero. Then, with a relatively loose matching criteria, track reconstruction becomes possible.

Next, the `AlignmentTrackChi2` module is used to perform automated and more precise alignment. Non-reference telescope planes are shifted along x,y directions and rotated about the z axis, and tracks are refitted with the trial geometries to minimise χ^2 values using Minuit2[34]. This process is performed iteratively until the alignment converges to a stable

[1] Note that the spatial correlations can be calculated without track information, so tracking is not required for prealignments.

configuration. The resulting geometry is written into an output file for subsequent analysis. However, it is possible that the angular orientations does not converge, due to the rotational symmetry of the setup and the telescope plane about z axis.

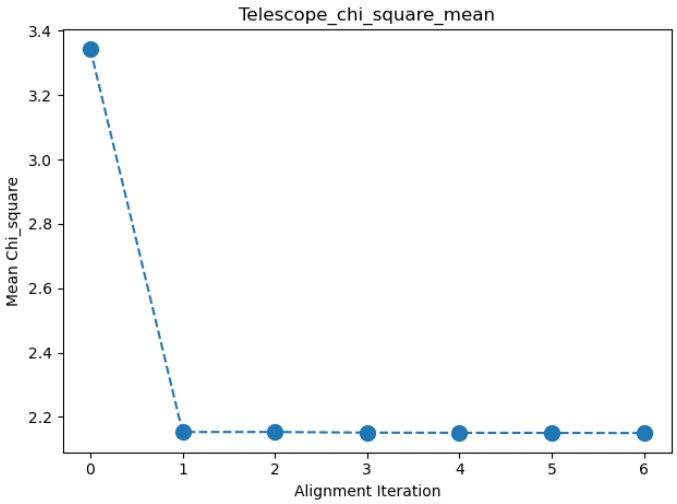


Fig. C.1 The change of mean χ^2 value of the tracks during telescope alignment.

The result of the alignment is checked with the x,y residuals calculated by the `AnalysisTelescope` module, where

$$x_{residual} = x_{cluster} - x_{track} \tag{C.2}$$

refers to the difference between the position of the associated cluster and the track intersection on telescope planes. Unlike correlations, residuals are track-dependent. The goal of a fine-tuned alignment is to minimise spatial residuals and centre their distributions at zero[2].

A similar procedure can be applied to the DUT by the `AlignmentDUTResidual` module, where only the DUT position is modified.

[2]In contrast, spatial correlation peaks may remain offset due to the inherent differences between the x,y position of detector planes.

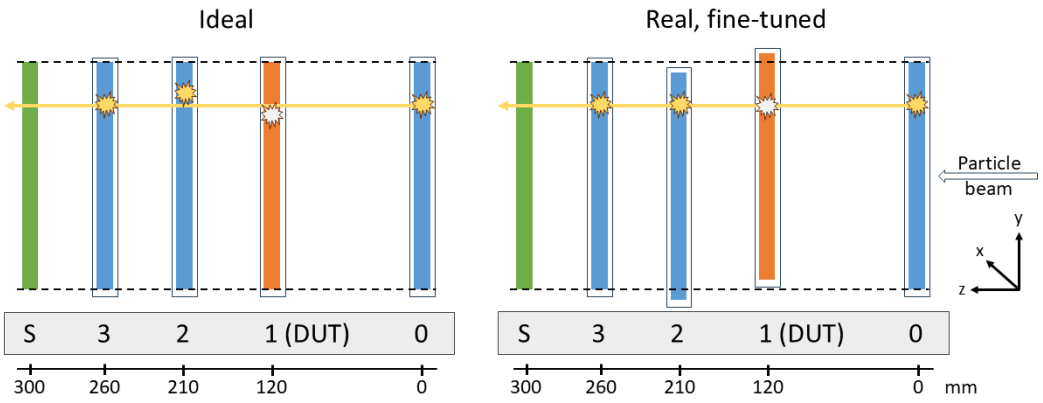


Fig. C.2 An illustration of a straight track fit through an (left) ideal and (right) fine-tuned, real alignment. Offsets are exaggerated for clarity. In the fine-tuned case, spatial residuals are centred around zero, indicating good alignment, while spatial correlations are offset due to difference in detector positioning.

Appendix D

Chip and Pixel Bias Voltage

There are two ways to apply bias voltage to the sensors: either by the chip guard ring or the pixel guard ring, as shown in Fig. D.1.

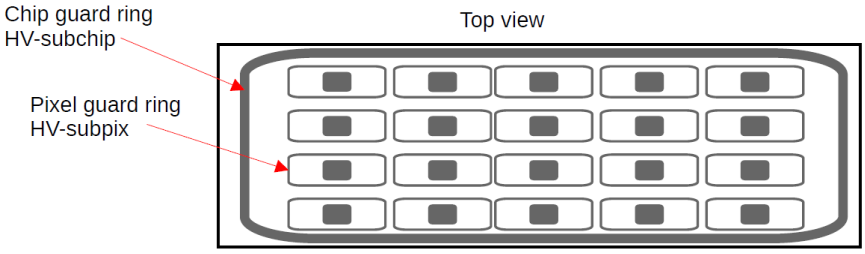
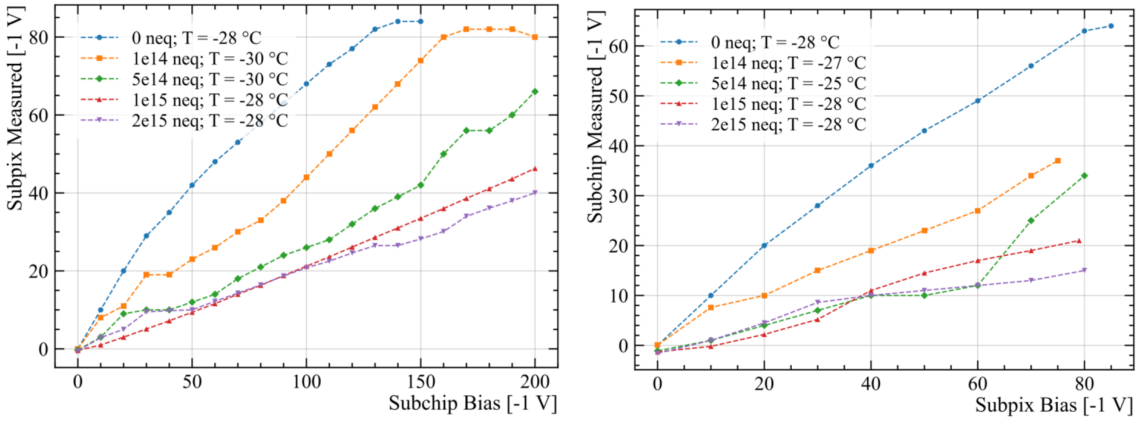


Fig. D.1 An simplified front view of the TelePix2 sensor[35]. Chip guard ring and pixel guard rings, from which the bias voltage can be applied.



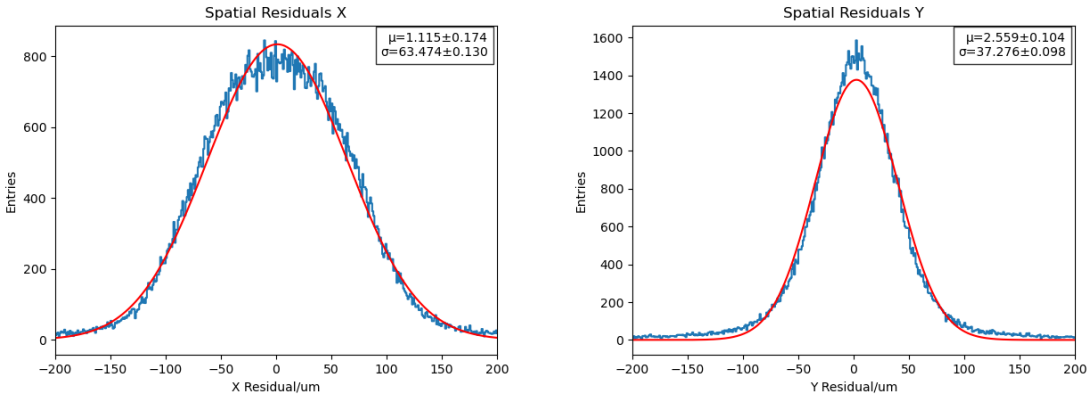
(a) Voltage applied on chip and measured on pixel. (b) Voltage applied on pixel and measured on chip.

Fig. D.2 The measured voltage on pixel guard ring when bias was applied on chip guard ring, and vice versa[35]. Note that the values are not one-to-one corresponded.

Appendix E

Spatial Resolutions of TelePix2

The spatial resolutions of TelePix2 were also estimated by fitting the spatial residual distributions with Gaussian function. The resolutions were measured to be $63.474 \pm 0.130 \mu m$ in x direction and $37.276 \pm 0.098 \mu m$.



(a) Plot of x residuals.

(b) Plot of y residuals.

Fig. E.1 Spatial residuals of TelePix2.